

Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

SUPPLEMENTARY APPENDIX

CRISPR-Based Therapy for Hereditary Angioedema

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DM, JSB, DLe, and HJL were involved in study conception and design. DMC, PG, MM, CHK, DLa, LB, RSP, KL, EAP, DM, MYS, AG, YX, AMA, and HJL were involved in data collection. DM, JSB, MYS, AG, YX, and AMA analyzed the data. DMC, PG, MM, CHK, DLa, LB, RSP, KL, EAP, and HJL were investigators for the study. The first draft of the manuscript was written by Justine Lempart, PhD, and James Banigan, PhD, employees of Apollo Medical Communications, part of the Helios Global Group (funded by Intellia Therapeutics), and MYS. All authors contributed to the data interpretation, review, editing, and approval of the manuscript for publication, and participated in the decision to submit for publication. All authors assume responsibility for the completeness and accuracy of the data, the fidelity of the trial and this manuscript, the protocol (available at NEJM.org), and the statistical analysis plan. All authors had access to all data in the study and agreed to maintain confidentiality of the data during manuscript development.

SUPPLEMENTARY METHODS

Screening

At screening, patients were asked, “What was your sex (at birth)?” Options were male or female. No patients identified as Hispanic or Latino. Patients were asked, “Do you consider yourself Hispanic/Latino or not Hispanic/Latino?” Options were Hispanic/Latino; not Hispanic/Latino; not reported; unknown. Twenty-six patients identified as White or Caucasian, and one patient identified as Native Hawaiian or other Pacific Islander. Patients were asked, “Which of the following designations best describe you (more than one choice is acceptable)?” Options were American Indian or Alaska Native; Asian; Black or African American; Native Hawaiian or other Pacific Islander; White or Caucasian; not reported; unknown.

Concomitant Medications

Patients were permitted to use on-demand medications to treat angioedema attacks during the study. The following medications were prohibited during the screening and 16-week primary observation period: kallikrein B1 (KLKB1) targeted therapies, including antisense oligonucleotides, prekallikrein, and/or kallikrein-lowering therapies; ecallantide; attenuated androgens and berotralstat; lanadelumab; anti-fibrinolytics; any other therapy intended for long-term prophylaxis (prevention) of hereditary angioedema (HAE) attacks; chronic antithrombotic therapy; ACE inhibitors; and estrogen-containing medications with systemic absorption. Following the 16-week primary observation period, patients were permitted to initiate other therapy to manage their HAE at the discretion of the investigator.

Supplementary Statistical Analysis

Pharmacodynamic analyses included all patients in the safety set in whom the total plasma kallikrein protein level was measured at least once before and at least once after the administration of study drug. Pharmacokinetic analyses included patients in the safety set who had at least one blood sample in which concentrations of any of the four plasma analytes of NTLA-2002 could be evaluated. Patient-reported outcome end point analyses were performed in the full analysis set. For angioedema attack-related end points, the 95% CIs were reported for model-based estimates, mean and standard error were presented for the descriptive summary; for all other end points, descriptive statistics (n, mean, median, standard deviation, minimum, maximum) were used in summarizing the data.

The primary end point of number of angioedema attacks per month during the 16-week primary observation period was compared between NTLA-2002 and placebo using a negative binominal regression model adjusting for baseline attack rate. The model included the treatment arm and the baseline attack rate as independent variables. The logarithm of follow-up duration (in months) for each patient during the primary observation period was included in the model as an offset variable to adjust for differences in follow-up time. From this model, the least squares mean attack rate and the corresponding 95% CI for each treatment arm as well as the mean rate ratio and the corresponding 95% CI for NTLA-2002 relative to placebo were estimated. These estimates were reported as mean angioedema attack rates per month by transforming the estimates using the exponential function. Missing data were assumed to be missing at random in this analysis. Patients who did not have a week 16 assessment at the time of the analysis had the time-adjusted attack rate included in the analysis. Pearson Chi-square test was performed to assess goodness of fit of the model, and the dispersion coefficient estimate from the model was examined for overdispersion of the data.

The angioedema attack-related secondary end points were analyzed using the same model approach as the primary end point. The angioedema attack rate and percentage change from baseline were also

summarized descriptively for all angioedema attack–related end points. Total plasma kallikrein protein level values and percentage change from baseline at each assessment were summarized descriptively. Absolute change in the AE-QoL scores from baseline at each scheduled post-baseline visit was summarized descriptively.

Additional details are in the statistical analysis plan.

SUPPLEMENTARY RESULTS

Pharmacokinetics

Following NTLA-2002 administration, all four analytes demonstrated a dose-proportional plasma exposure (**Fig. S4**). NTLA-2002 analytes rapidly declined from peak concentrations, followed by a secondary peak at approximately 12 hours post infusion, and then a log-linear elimination phase. All analytes, except DMG-PEG2k (biodegradable by design)¹ were quantifiable in plasma up to 29 days post infusion. DMG-PEG2k declined to below quantification limits within approximately 24 hours post end of infusion.

Phase 1 Update

Baseline characteristics were reported previously.² In brief, the median age was 51 (range, 26 to 73) years, 60% were male, and the median number of angioedema attacks during the 90 days before screening was 6 (range, 3 to 54).

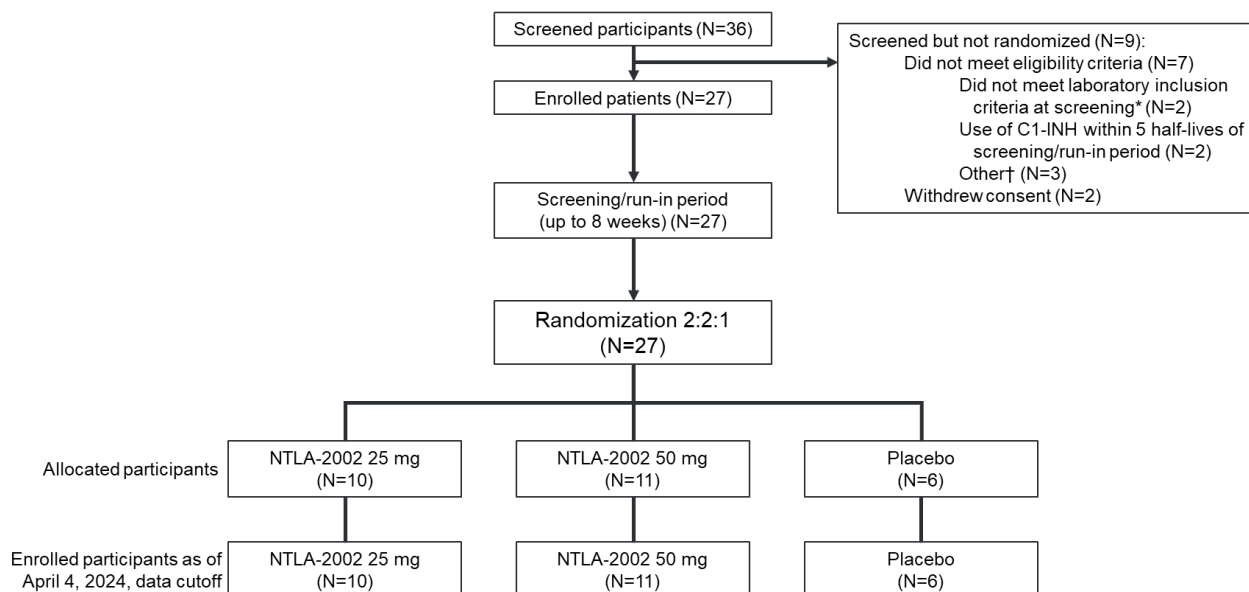
Between the end of the primary 16-week observation period and the latest assessment (week 104, 25 mg; week 72, 50 mg; week 88, 75 mg) the mean percentage change from baseline in monthly angioedema attack rate was –99%, –100%, and –100% in the 25-mg, 50-mg, and 75-mg arms (**Table S5, Fig. S6**). The mean percentage change in monthly angioedema attack rate between baseline and the latest assessment was –97%, –100%, and –97% in the 25-mg, 50-mg, and 75-mg arms. Eight patients remain attack-free since the end of the primary observation period (**Fig. S7**). None of the six patients who stopped their long-term prophylaxis after administration of NTLA-2002 have resumed, and five reported no attacks since withdrawal.

NTLA-2002-treated patients continued to show a low rate of adverse events (AEs) (**Table S6**). No dose-limiting toxic effects, serious AEs, or AEs of grade 3 or higher occurred after administration of NTLA-2002. The most common AEs were infusion-related reactions (70%), COVID-19 (60%), fatigue (60%), and upper respiratory tract infection (50%).

Dose-dependent reductions in the total plasma kallikrein protein levels were sustained, with a mean (SD) percentage change from baseline to the latest assessment of –70% (21.4), –88% (6.8), and –95% (1.1) in the 25-mg, 50-mg, and 75-mg arms, respectively (**Fig. S8**).

Figure S1. CONSORT Diagram.

Flowchart of screening, enrollment, and allocation of patients in the phase 2 portion of the phase 1–2 study (NCT05120830) of NTLA-2002 vs. placebo in patients with hereditary angioedema.



*Patients were required to meet laboratory inclusion criteria during screening, including aspartate aminotransferase, alanine aminotransferase, and total bilirubin less than or equal to the upper limit of normal range.

†One patient was exposed to angiotensin-converting enzyme inhibitors or any estrogen-containing medications with systemic absorption within 90 days prior to study drug administration. One patient used antithrombotic therapy (other than aspirin) within 14 days prior to study drug administration. One patient had laboratory abnormality, psycho-social stressor, pattern of behavior, or other reason that, in the Investigator's opinion, could adversely affect the safety of the subject, impair the assessment of study results, or preclude compliance with the study.

Figure S2. Mean Number of Investigator-Confirmed Angioedema Attacks per Month over Time.

The monthly angioedema attack rate is calculated as the number of angioedema attacks that occurred during each 28-day interval, with the date of NTLA-2002 or placebo administration as the start of the first interval. Error bars indicate standard error. Patients were required to withdraw long-term prophylaxis prior to study entry but were permitted to resume any long-term prophylaxis after the end of the primary observation period at week 16 (month 4). Re-introduction of long-term prophylaxis in four of the five patients in the placebo arm who had reached the week 16 visit is likely contributing to the decrease in the attack rate after month 4 in the placebo arm. BL, baseline; HAE, hereditary angioedema.

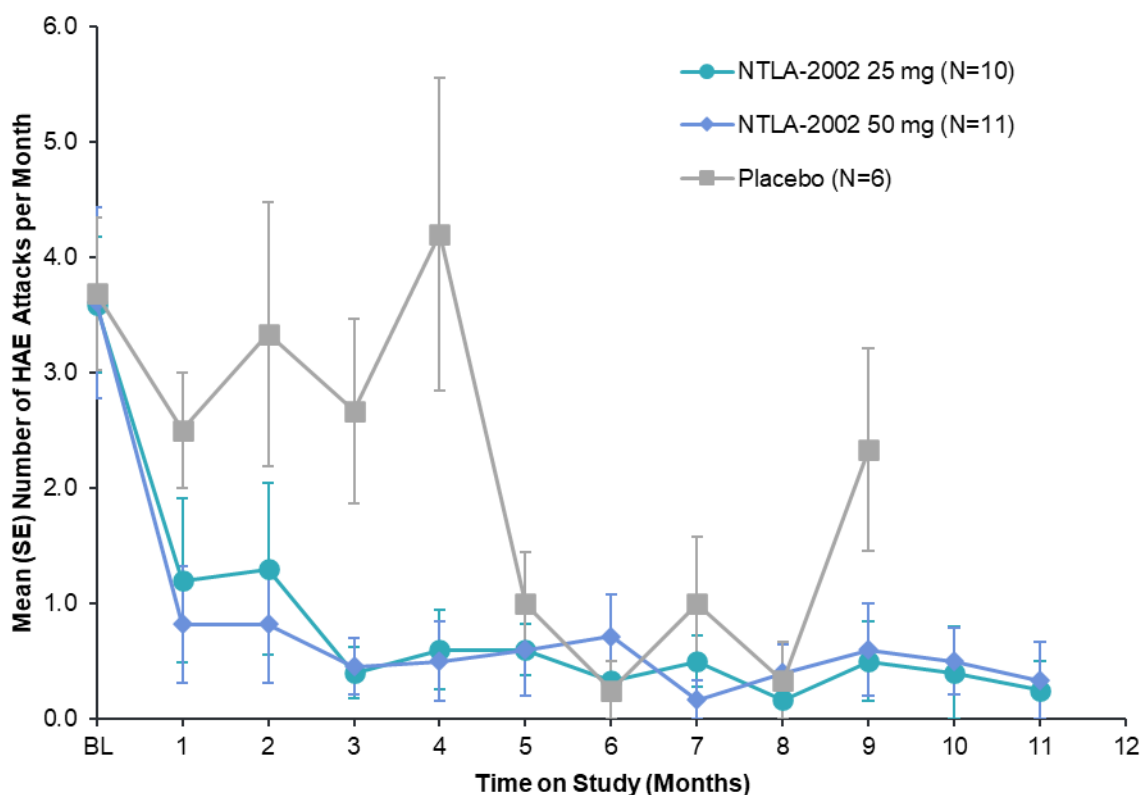


Figure S3. Responder Status.

Investigator-confirmed angioedema attack response status during the 16-week primary observation period. Response status is defined as the patients’ monthly attack rate reduction from baseline of at least 50%, 70%, 90%, and 100% (attack-free). HAE, hereditary angioedema.

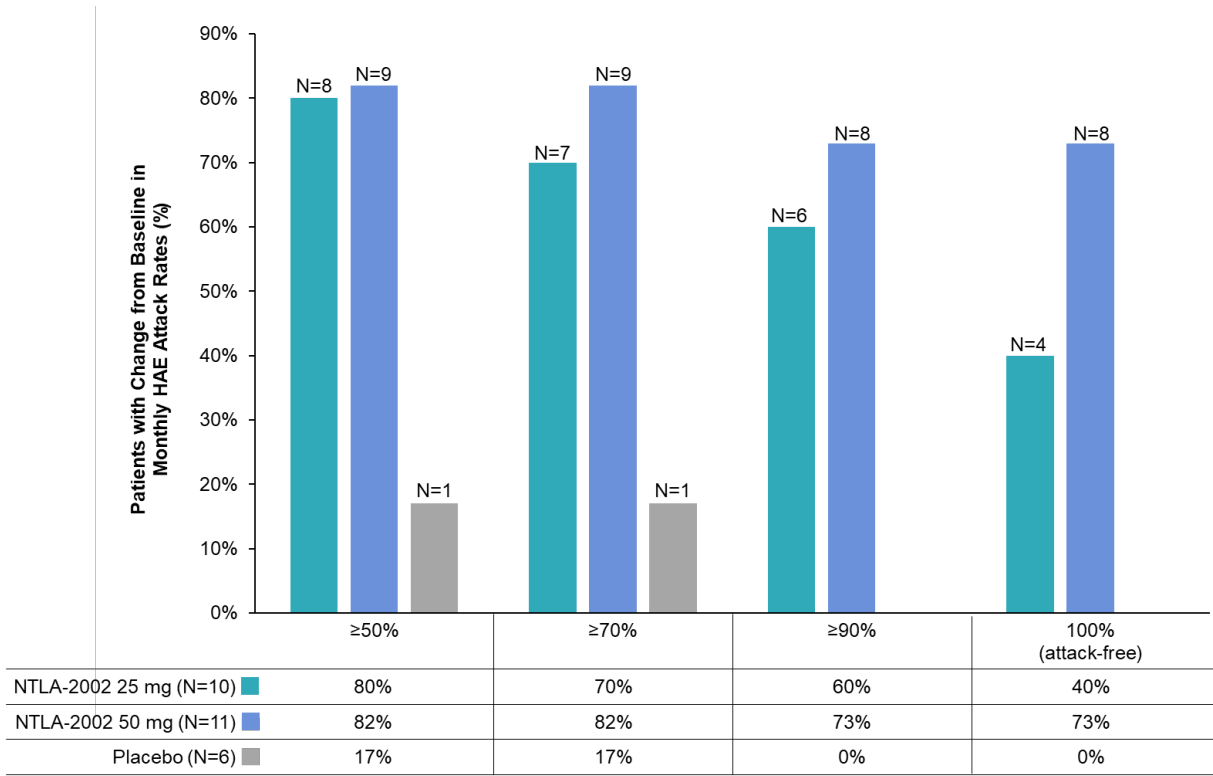


Figure S4. Pharmacokinetics of NTLA-2002 Analytes.

Data shown are mean geometric values (semi-log scale) of NTLA-2002 analytes post end of NTLA-2002 infusion: (A) DMG-PEG2k, (B) LP000001, (C) Cas9 mRNA, and (D) *KLKB1* sgRNA.

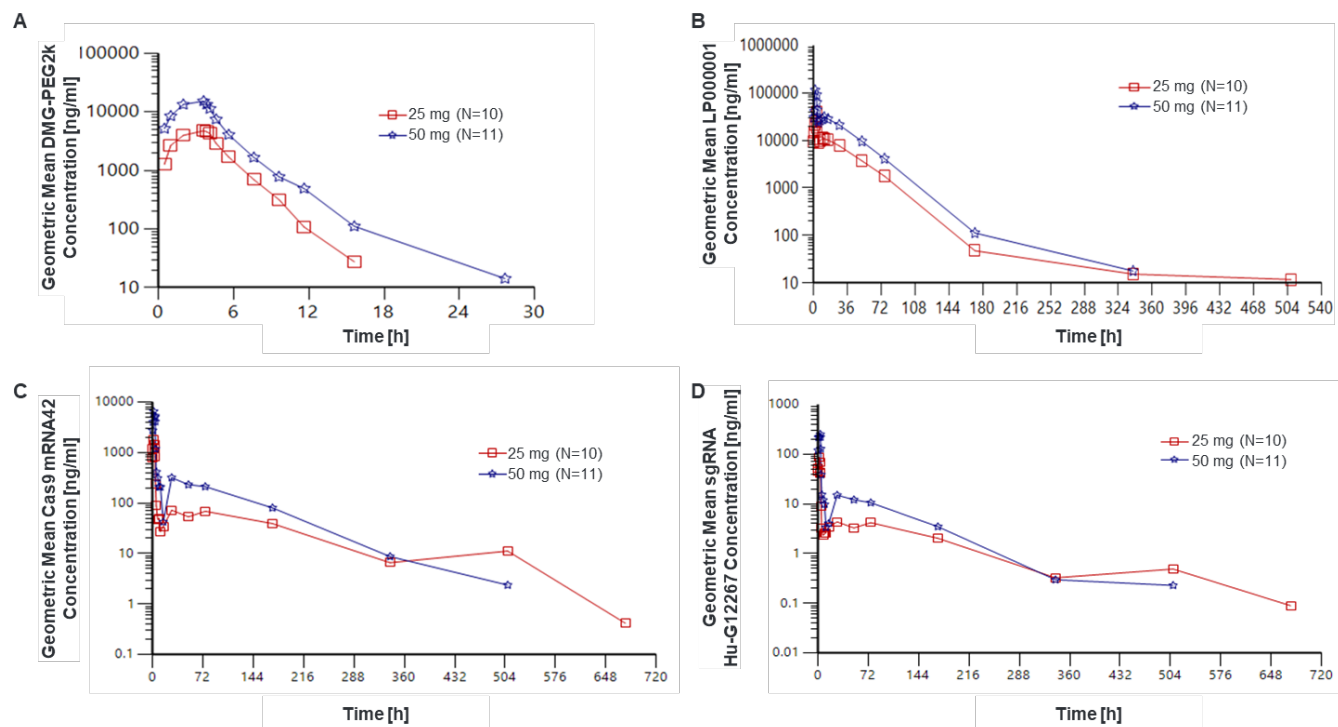


Figure S5. Mean Change in AE-QoL Scores.

Data shown are the mean (SD) change from baseline at week 16 in the total AE-QoL score and AE-QoL scores of functioning, fatigue/mood, fear/shame, and nutrition. Only patients with both baseline and week 16 AE-QoL assessments were included in the analysis. The dashed line represents the total score minimal clinically important difference of 6 points.³

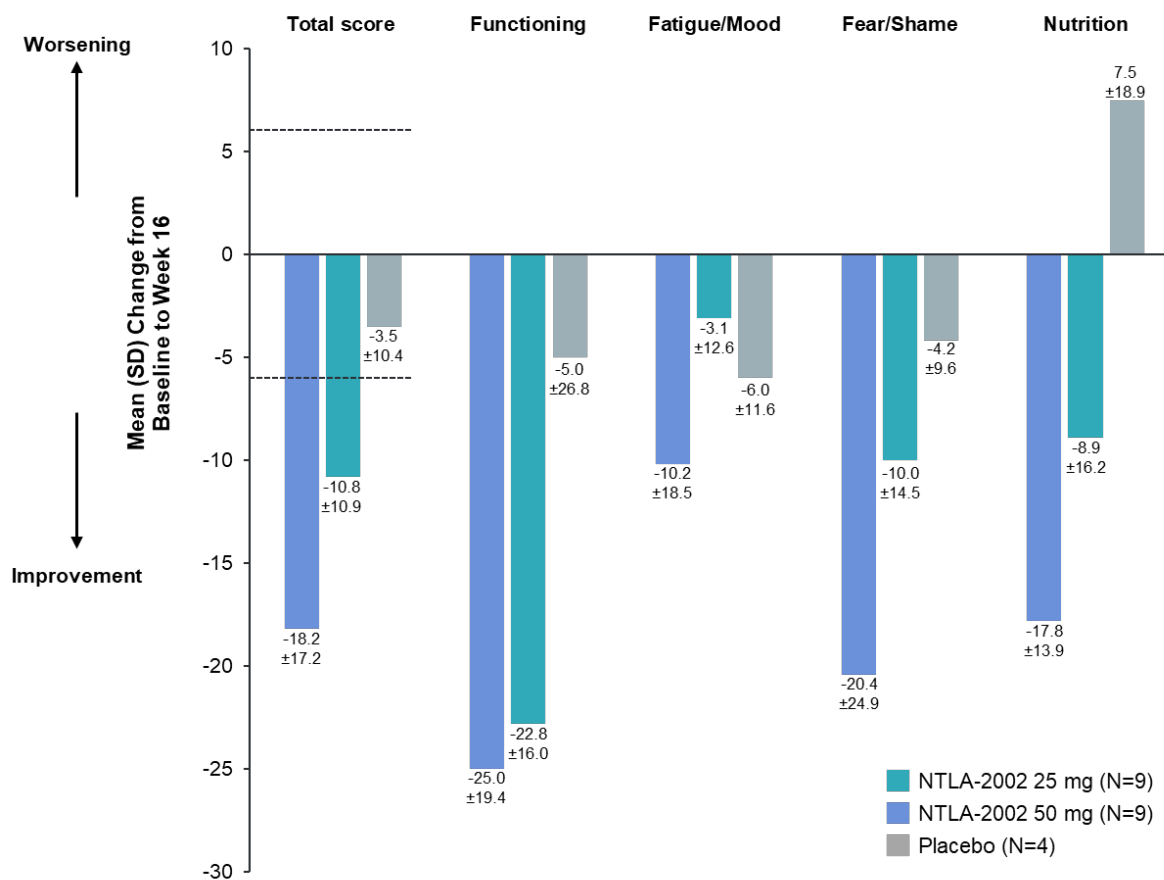


Figure S6. Mean Number of Investigator-Confirmed Angioedema Attacks per Month over Time in the Phase 1 Study.

The monthly angioedema attack rate is calculated as the number of angioedema attacks that occurred during each 28-day interval, with the date of NTLA-2002 administration as the start of the first interval. Error bars indicate SD. Patients were permitted to use concomitant long-term prophylaxis and withdraw it at their discretion.² BL, baseline, HAE, hereditary angioedema.

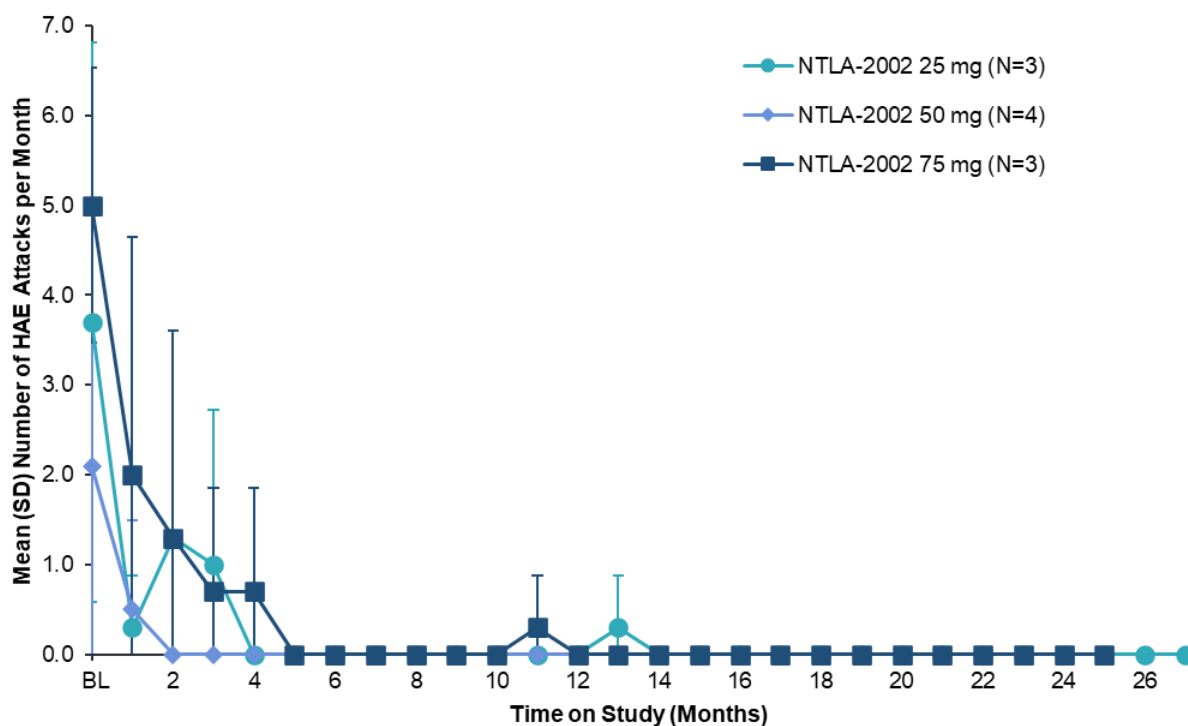


Figure S7. Angioedema Attacks before and after the Infusion of NTLA-2002 in the Phase 1 Study.

Each row shows individual patient data on angioedema attacks; the number of attacks per month in each patient during the historical attack period (defined as the 90-day period before the screening period) is also shown. Patient data are listed from top to bottom according to the length of follow-up. The length of the gray bars indicates the interval from the start of the screening period to the last assessment as of the data cutoff date. Arrows indicate that the patient is still on study. The width of the colored bars indicates the duration of an attack. Black triangles indicate the time of withdrawal of any concomitant long-term prophylaxis. Patients without black triangles were not receiving concomitant long-term prophylaxis.

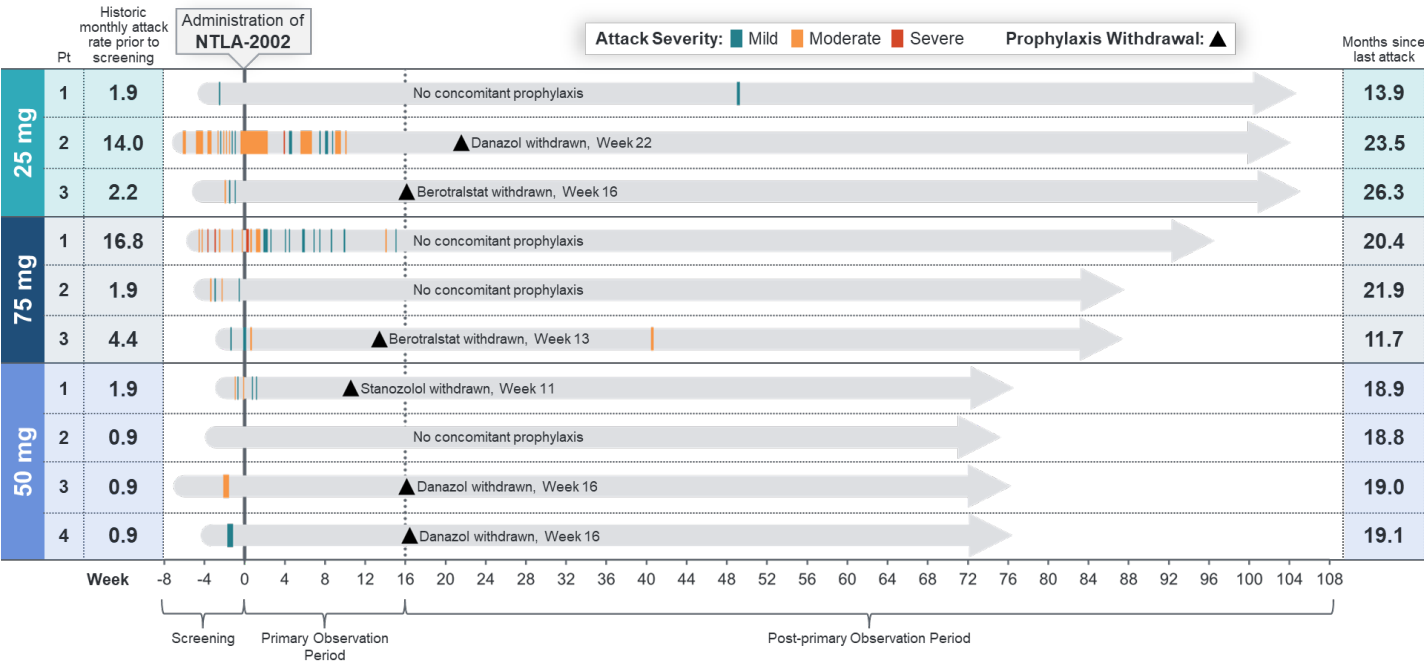


Figure S8. Change from Baseline in Total Plasma Kallikrein Protein Levels over Time by Dose Level in the Phase 1 Study.

Shown is the mean percentage change from baseline in the total plasma kallikrein protein level in each dose cohort. Error bars indicate SD. The dashed line represents the targeted minimum reduction. Only visits completed by all patients within a cohort are presented. BL, baseline.

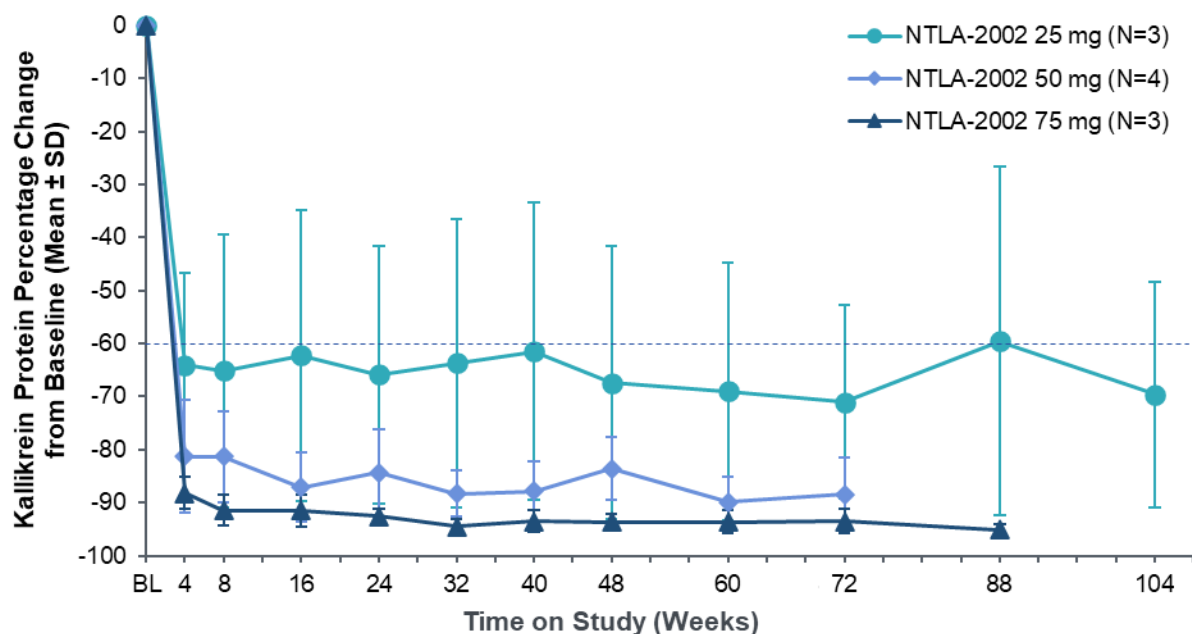


Table S1. Representativeness of Study Patients.

Category	
Disease under investigation	Hereditary angioedema due to C1INH deficiency (HAE-C1INH)
Special considerations related to:	
Sex and gender	Males and females are affected equally at the genetic level. Symptoms are sometimes more severe in female patients, possibly due to fluctuation in female sex hormones, and this may contribute to a slightly higher likelihood of diagnosis.
Age	HAE is an autosomal dominant genetic disease; therefore, all patients who are affected have the disease from birth. The age at which attacks begin is variable, but in most patients, the disease presents in childhood or adolescence. Patients will continue to have attacks throughout their lifetime.
Race or ethnic group	There are no known differences in prevalence, disease pathophysiology, or responses to treatment among racial or ethnic groups.
Geography	There are no known differences in prevalence in different countries. However, HAE is a rare disease (1:50,000 population) but shares symptoms with other more common diseases. This may contribute to patients being misdiagnosed or underdiagnosed in developing countries or countries that lack access to diagnostic tests and evidence-based therapies.
Other considerations	There are two main types of HAE: HAE-C1INH-Type 1 ($\approx 85\%$ of patients) and HAE-C1INH-Type 2 ($\approx 15\%$ of patients). Attacks most often affect the skin, gastrointestinal tract, and upper airway. The frequency and severity of attacks is highly variable across patients.
Overall representativeness of this study	This was a small phase 2 study of 27 adult patients with HAE-C1INH. The age distribution of study patients was consistent with what is observed in adults with HAE-C1INH. There was a slightly higher number of men than women. Participants were representative of patient populations in countries with study sites, which have predominantly White or Caucasian populations: Australia, France, Germany, the Netherlands, New Zealand, and the United Kingdom. The number of patients with HAE-C1INH-Type 1 and HAE-C1INH-Type 2 disease was representative of the overall HAE-C1INH population. Patients also had varying types, frequency, and severity of HAE attacks, which is typical in patients with HAE.

Table S2. Change from Baseline in Number of Investigator-Confirmed Attacks per Month.*

Variable	NTLA-2002 25 mg (N=10)	NTLA-2002 50 mg (N=11)	Placebo (N=6)
Percentage change from baseline in no. of attacks per month			
Weeks 1 to 16	-78.1±30.4	-79.5±42.0	-16.3±41.6
Weeks 5 to 16	-79.9±28.9	-81.3±38.8	-3.4±55.8
Percentage change from baseline in no. of attacks per month requiring on-demand treatment			
Weeks 1 to 16	-79.7±27.2	-73.8±63.7	-15.2±46.9
Weeks 5 to 16	-72.8±28.3	-74.0±63.3	-11.7±58.5
Percentage change from baseline in no. of moderate to severe attacks per month			
Weeks 1 to 16	-73.9±42.7	-82.8±35.6	-20.7±57.8
Weeks 5 to 16	-72.8±49.7	-87.6±26.4	-19.6±63.4

*Plus-minus values denote means ±SD. Baseline was defined as the time from the date of informed consent to randomization.

Table S3. Mean Number of Investigator-Confirmed Attacks per Month Requiring Acute Therapy or of Moderate to Severe Intensity.

Variable	NTLA-2002 25 mg (N=10)	NTLA-2002 50 mg (N=11)	Placebo (N=6)
Angioedema attacks requiring acute therapy			
Weeks 1 to 16			
Mean no. of angioedema attacks per month (95% CI)*	0.63 (0.21, 1.94)	0.55 (0.18, 1.66)	2.39 (0.60, 9.44)
Percentage difference vs. placebo (95% CI)	-74% (-95%, 51%)	-77% (-96%, 39%)	
Weeks 5 to 16			
Mean no. of angioedema attacks per month (95% CI)*	0.60 (0.18, 1.97)	0.53 (0.16, 1.72)	2.58 (0.61, 10.88)
Percentage difference vs. placebo (95% CI)	-77% (-96%, 46%)	-79% (-97%, 37%)	
Moderate to severe angioedema attacks			
Weeks 1 to 16			
Mean no. of angioedema attacks per month (95% CI)*	0.45 (0.14, 1.39)	0.32 (0.10, 0.97)	2.05 (0.54, 7.87)
Percentage difference vs. placebo (95% CI)	-78% (-96%, 26%)	-85% (-97%, -11%)	
Weeks 5 to 16			
Mean no. of angioedema attacks per month (95% CI)*	0.41 (0.11, 1.56)	0.23 (0.06, 0.91)	2.12 (0.45, 9.87)
Percentage difference vs. placebo (95% CI)	-80% (-97%, 50%)	-89% (-99%, -14%)	

*The mean number of angioedema attacks per month was estimated using a negative binomial model with treatment arm and baseline attack rate as independent variables. Baseline is defined as the time from date of informed consent to randomization. A month is defined as 28 days.

Table S4. Summary of Infusion-Related Reactions.

Event — no. (%)	NTLA-2002 25 mg (N=10)	NTLA-2002 50 mg (N=11)	Placebo (N=6)
Any event	1 (10)	3 (27)	1 (17)
Back pain	1 (10)	1 (9)	0
Flushing	0	1 (9)	1 (17)
Other*	0	3 (27)	0

*Other included chest pain (noncardiac), generalized tingling, neck pain, and tight chest; tight chest and generalized tingling were experienced by the same patient.

Table S5. Change from Baseline in Number of Investigator-Confirmed Attacks per Month in the Phase 1 Study.*

Variable	NTLA-2002 25 mg (N=3)	NTLA-2002 50 mg (N=4)	NTLA-2002 75 mg (N=3)	All Patients (N=10)
Patients with attacks during the screening period — no. (%)	3 (100)	3 (75)	3 (100)	9 (90)
Percentage change from baseline in no. of attacks per month†				
Post-primary observation period	-98.7±2.3	-100±0.0	-99.6±0.8	-99.4±1.4
On-study period‡	-97.4±2.3	-99.5±0.93	-96.6±4.1	-97.8±2.7

*Plus-minus values denote means ±SD. Baseline was defined as the screening period (≤42 days before infusion) before the administration of NTLA-2002.

†The percentage change from baseline was calculated only for patients with angioedema attacks during the screening period.

‡The on-study period was defined as the time from the administration of study intervention through the last assessment as of the data cutoff.

Table S6. Adverse Events after the Administration of NTLA-2002 in the Phase 1 Study.*

Event	NTLA-2002 25 mg (N=3)	NTLA-2002 50 mg (N=4)	NTLA-2002 75 mg (N=3)	All Patients (N=10)
<i>number of patients (percent)</i>				
Any adverse event	3 (100)	4 (100)	3 (100)	10 (100)
Infusion-related reaction	2 (67)	2 (50)	3 (100)	7 (70)
COVID-19	3 (100)	1 (25)	2 (67)	6 (60)
Fatigue	1 (33)	3 (75)	2 (67)	6 (60)
Upper respiratory tract infection	1 (33)	1 (25)	3 (100)	5 (50)
Myalgia	0	0	3 (100)	3 (30)
Oropharyngeal pain	2 (67)	0	1 (33)	3 (30)
Abdominal discomfort	0	2 (50)	0	2 (20)
Headache	0	0	2 (67)	2 (20)
Viral upper respiratory tract infection	0	0	2 (67)	2 (20)

*Shown are adverse events, coded according to preferred terms in the *Medical Dictionary for Regulatory Activities*, version 24.0, that occurred in at least two patients after the administration of NTLA-2002. No adverse events after the administration of NTLA-2002 were assessed to be grade 3 or higher in severity.

Supplementary References

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